



Evaluation of control methods for drilling operations with unexpected gas influx

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ABSTRACT

This paper presents an evaluation of various control methods to be used during drilling operations where an unexpected gas influx occurs. In the event of an unexpected gas influx the current industry procedure is to control the pressure in the well manually. The drilling industry term for this manual procedure is *well control*. The focus of the paper is threefold. Firstly, to design an automatic sequence which is similar to the existing manual procedure. Secondly, to evaluate three different control algorithms for pressure control during an unexpected gas influx, and thirdly, to evaluate control parameter tuning needed when implementing different control algorithms.

The control methods have been evaluated on various drilling scenarios with unexpected gas influx, referred to as a *kick*. After a kick of reservoir gas has entered the well, automatic control of the well control choke and rig pump is applied to compensate for pressure fluctuations while circulating out the gas. A PI controller is designed to stabilize the well pressure by controlling the well control choke, an internal model controller (IMC) controls the pressure by manipulating the choke and the rig pump flowrate, and a model predictive controller (MPC) uses coordinated control of the choke and the pump flowrate to stabilize the well pressure. The model based controllers use a simple first order model of the well. Simulations are performed using a detailed flow model of the well to test the controller performance and robustness. Several cases with different amounts of gas influx are investigated.

The simulations show that it is feasible to control the pressure using automatic control of the choke valve and pump during an unexpected gas influx by use of all the presented control methods. The control methods are robust against changes in process conditions and disturbances, as they are able to handle several pressure levels and gas volumes without requiring re-tuning. However, since the pressure dynamics in the well are influenced when gas is entering the well, the model based controllers could probably be further improved if the models were updated after the gas influx occurred.

The results indicate that adaption of the automatic sequence to the current manual procedure is applicable. However, to avoid a reduction in downhole pressure when stopping the pump and shutting in the well, the automatic sequence may be further improved beyond what is feasible with manual operation.

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1. Introduction

While drilling in the reservoir section of a well, pressure control is crucial. The well pressure in this section should stay above the pore pressure of the formation. If the pressure falls below the pore pressure, reservoir fluids or gas will leak into the well. This is called an *influx*, and if the influx is above a certain size it is termed a *kick*. If not handled properly, this may lead to a blowout situation, where

the reservoir fluids flow uncontrolled into the wellbore and to the surface. On the other hand, if the pressure in the well is too high, this may cause loss of mud to the formation and well fracture.

A kick can be caused by too low mud density, such that the hydrostatic pressure in the well is below the pore pressure, or by transient effects like drillstring movement. There may be several indicators that a kick has occurred. One indicator is increased volume in the pit tank, which is the tank containing the drilling mud. If gas starts to leak into the well, the volume of the gas will occupy the annulus volume and push the drilling fluid upwards. The flow rate out of the annulus will then be higher than the flow rate pumped into the drillstring from the pit, causing the volume in the pit tank to increase. A quicker indicator is comparing the flow rate into and out of the well directly, not through the pit volume. This however requires a flow meter at the outlet of the well, which is usually

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not available. Other kick indications are sudden increase in drilling rate, change in pump pressure and reduction in drillpipe weight [1].

If the kick is above a certain size, the well needs to be shut in to stop the kick. The kick must then be circulated out in a controlled manner. This procedure, called *well control*, is a manual operation including sensor recordings, calculations and control performed by several members of the drilling crew. It involves control of the blow-out preventer (BOP), the rig pump and the well control choke, all located at different locations at the rig. The well control choke is adjusted manually in order to maintain a certain pressure in the well. This may be a difficult task due to large time-delays in the drilling process and the complex behavior of the multiphase flow. Automation of the well control procedure including pressure control with automatic choke adjustments would improve reliability and reduce the possibility of human errors.

The lack of automation level within the industry may have several reasons: firstly, the drilling operation is performed partly by the oil company, the drilling contractor and the service company. This means that the overall process overview is not fully understood by each of the companies involved in the drilling operation. This applies specifically to the service companies since they typically deliver equipment and personnel trained to operate their equipment. Because automation would mean less personnel involved, this would lead to reduced revenue for the service companies.

In other industries automation has evolved due to economic reasons, and since oil drilling has not reached sufficiently hard cost constraints no motivation for automation has been present. However, since automation also gives improved Health, Safety and Environment (HSE), new focus on automation of the drilling process can currently be seen.

In recent years, several drilling operations have been performed successfully using managed pressure drilling (MPD) techniques. Managed pressure drilling is a collective term for various techniques employed to manage the annular hydraulic pressure profile of the exposed wellbore [2]. A typical layout of an annulus back-pressure type MPD system is described in [3,4], and control requirements for MPD systems are discussed in [5]. In [6,5] PID control methods are applied to control the pressure during MPD operations. Nonlinear model based control solutions and observers are addressed in [7] together with experimental results. The use of model predictive control is proposed as a multivariable control framework for dual-gradient drilling by [8] and for an annulus back-pressure systems by [9]. A multilevel control approach of an MPD operation is presented in [10,11].

With managed pressure drilling systems, a new possibility of handling small kicks without shutting in the well and increasing the mud weight to increase the hydrostatic pressure is introduced. Instead, the influx can be stopped by increasing the well pressure through adjustment of the MPD choke. Several papers describe the need for new well control procedures during drilling operations utilizing equipment designed for managed pressure drilling including both annulus backpressure systems and dual gradient systems [7]. An evaluation of different approaches of initial responses to kick during MPD operations is given by [12]. For annulus back-pressure systems, a well control method described as micro-flux control is described by [13] where the difference between flow in and flow out of the well is compared, and instant choke adjustments are made to maintain control over the well. An evaluation of well control procedures for dual gradient systems including riser-gas lift is given by [14]. In [15], a kick detection and attenuation algorithm is developed for a dual gradient system. Managed pressure drilling also introduces new techniques for kick detection and pore pressure estimation as described by [16,17].

The new techniques for kick handling during MPD operations are only applicable for relatively small kicks. When experiencing

large kicks, it is still recommended to shut in the well and circulate out the kick using standard well control procedures and equipment. Only a very low percentage of wells today are drilled using MPD techniques. In most kick situations MPD equipment is not available, and standard well control procedures using the well control equipment must be applied.

In this paper, ideas and pressure control methods from MPD operations are applied to automatically control the pressure while performing standard well control procedures with the well control equipment. It is demonstrated that simple control methods can be applied for the purpose. In the event of a larger kick event the standard industry procedure today is still to control the pressure in the well manually. The focus of the paper is threefold. Firstly, to design an automatic sequence which is similar to the existing manual procedure. Secondly, to evaluate three different control algorithms for pressure control during an unexpected gas influx, and thirdly, to evaluate control parameter tuning needed when implementing different control algorithms.

Section 2 of this paper gives an overview of a conventional drilling system and describes how standard well control procedures are performed manually today. In Section 3, the details for the well test case used during the simulations are given. A first-order approximation model of the system to be used in the controller design is presented. Section 4 presents three different control methods for controlling the pressure during well control incidents, a PI control design, an internal model control algorithm and a model predictive controller. The controllers are evaluated by running different simulation scenarios on a high-fidelity simulator in Section 5. The next section discusses the tuning of parameters of the various controllers, and conclusions and ideas for further work are given in Section 7.

2. Drilling and well control

A schematic of a conventional drilling system is shown in Fig. 1. The system consists of a rotating drillstring with a drilling bit at the bottom. The volume that develops around the drillstring as the well is drilled is referred to as the annulus. Drilling mud is pumped from a pit tank, down the drillstring, through the bit, up the annulus, and back into the pit tank. The purpose of the drilling mud is to transport cuttings from the bottom of the well, and to create a certain pressure balance in the well. The pressure in the well during a conventional drilling operation is managed by the density and the flow rate of the drilling fluid.

The well control equipment is depicted in Fig. 1. A blow-out preventer is located on the top of the annulus, or at the sea floor on floating rigs. The purpose of the BOP is to seal the annulus in a kick situation, to prevent reservoir fluids from flowing uncontrolled out of the well. When the BOP is closed, the circulation path is directed through the choke line, located below the BOP, through the well control choke, through the de-gasser, and into the pit tank.

During a conventional drilling operation, the only secure method to stop a kick is to shut in the well. A typical shut-in procedure [1] is to stop the pumps, open the well control choke, close the blow-out preventer, and close the well control choke. This is termed a “soft shut-in”. The alternative is to close the blow-out preventer on a closed choke, which is known as a “hard shut-in”.

When the well has been shut in, the pressure in the well will start to increase. This is because as long as the well pressure at the location of the influx is below the pore pressure, gas will continue to flow into the well, filling the constant volume in the closed annulus, which will cause the pressure increase. When the pressure at the influx location has increased above the pore pressure, the influx stops and the pressure stabilizes.

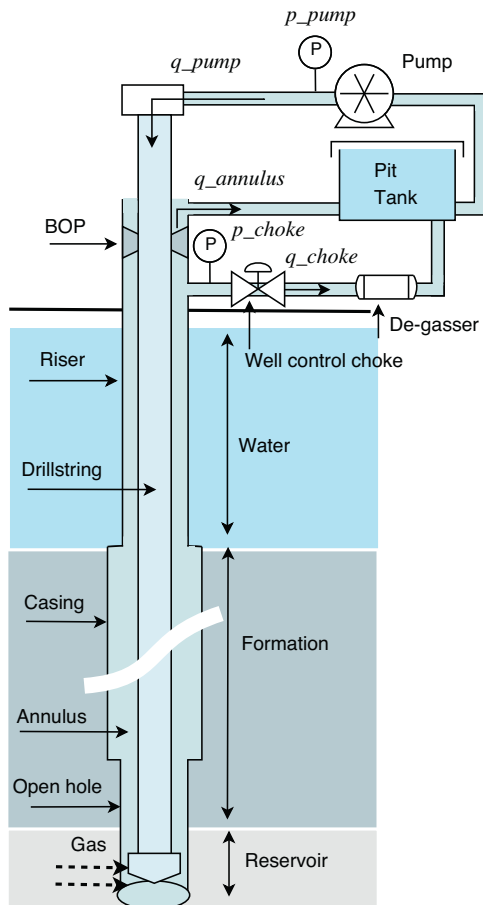


Fig. 1. Drilling system.

Once the influx is stopped and the pressure is stable, the procedure continues by circulating out the influx that has entered the annulus. The influx is circulated out through the choke line below the blow-out preventer. While circulating out a gas kick, the gas must be allowed to expand to avoid rupturing the wellbore. At the same time, the pressure at the bottom of the hole must be kept above the pore pressure to prevent additional influx from the reservoir. The goal of classic well control procedures is to maintain a constant downhole pressure while circulating out the kick. The pressure is controlled by adjusting the well control choke at the outlet. However, measurements of the downhole pressure are usually not available. Only topside measurements such as pump pressure, also referred to as the drillpipe or standpipe pressure, and measurements of the pressure at the inlet of the well control choke, termed choke or casing pressure, can be used.

Classic well control procedures apply these pressure measurements to indirectly control the downhole pressure. These procedures are based on the assumption that when using slow pump rates, the frictional pressure drop in the annulus can be neglected, focusing on frictional pressure loss in the drillpipe only.

When the pump is running at constant rate, the pressure at downhole will remain constant if the pump pressure is constant. But when ramping up or down the pump, keeping the pump pressure constant will not give a constant downhole pressure due to frictional pressure changes in the drillpipe.

During pump ramp up and ramp down the downhole pressure will remain approximately constant if the choke pressure is kept constant. This is however not the case in wells with a large frictional pressure drop in the annulus, for example in slim holes. When starting up the pump in wells with large frictional pressure drop

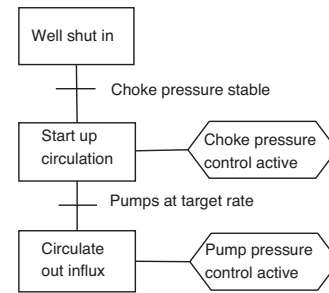


Fig. 2. Well control.

in the annulus, keeping the casing pressure constant will cause an increase in the downhole pressure. But if the pressure increase is not too high, it will simply be a safety barrier towards the pore pressure.

The most applied well control procedures are the Driller's Method and the Wait and Weight Method [1]. The Driller's Method is simple and requires minimal calculations. The procedure, depicted in Fig. 2, starts by bringing the pump to a desired circulation rate, keeping the casing pressure constant at the shut-in casing pressure, which is the casing pressure when the pressure has stabilized after a shut-in. When the pumps are running at the desired rate, the influx is displaced while keeping the drillpipe pressure constant. In order for the well to regain overbalance, the mud weight must be increased. This is referred to as to *kill* the well. The well is in overbalance when the hydrostatic pressure caused by the drilling fluid in the wellbore is higher than the pore pressure. The Driller's Method continues by circulating the heavier kill-weight mud through the drillpipe and finally displacing the heavier mud to the surface. The well should then be in overbalance, and it is safe to open the blow-out preventer and continue the drilling operation.

2.1. Instrumentation

Instrumentation during drilling operations is poor. The pressure at the bit is sometimes measured, but the signal is sent with mud pulses to the surface, which limits bandwidth significantly. The signal will thus have a delay of up to several seconds and cannot be utilized for control purposes. The pump pressure p_{pump} is measured with a pressure sensor placed along the path between the mud pump and the drillpipe, as depicted in Fig. 1. In a well control situation, when the circulation path goes through the well control choke, the choke pressure p_{choke} is measured with a pressure sensor placed before the choke, also depicted in Fig. 1. The pump flow rate, q_{pump} , is measured by counting the pump strokes per minute. When the volume of each stroke is known, the pump strokes per minute can be converted to volume flow rate per minute. Unknown pump efficiency and pump leakage can cause incorrect flow rate readings. The flow rate out of the well, $q_{annulus}$, is measured using a flow paddle placed in the return channel from the annulus to the pit tank. This is an inaccurate flow measurement.

3. Case description

3.1. Simulation model

The simulations performed in this paper are evaluated using a detailed multiphase fluid model [18], modeling both the wellbore dynamic pressure and velocity effects, in addition to the transient behavior of the influx from the reservoir. The model assumes dispersed bubble flow and calculations are performed using a drift-flux model. Temperature effects are included in the model. However, the temperature interaction between the well fluids and

Table 1
Well and reservoir data.

Parameter	Value
Well total length	2300 m
Well vertical depth	1720 m
Reservoir height	2 m
Reservoir permeability	100 mD
Reservoir pore pressure	262 bar
Reservoir porosity	0.18
Skin factor	0.13
Fluid density	1.475 s.g.

formation fluids are not handled. Reference to the model is given in [9]. The simulation model has been verified by several real-time applications [19,20].

An inclined well of length 2300 m is used as a test case, with a water based drilling fluid with density 1475 kg/m³. Further details of the well, the drilling fluid and the reservoir conditions are given in Table 1.

3.2. Control objectives and variables

The control objective is to keep the pressure at the bottom of the well constant during a well control situation. When measurements of the downhole pressure are available, the downhole pressure can be controlled directly. When these measurements are not available, the control objective is to keep the choke pressure constant while starting up the pump, and keeping the pump pressure constant while circulating out the gas. The main manipulated variable is the choke opening of the well control choke, situated at the outlet of the well as depicted in Fig. 1. The internal model control (IMC) method and the model predictive control (MPC) method used in this paper have the pump flow rate as a second manipulated variable. For the PI controller, the pump flow rate will be a disturbance. Other unmeasured disturbances are gas flow and expansion in the annulus and through the choke, which will cause pressure fluctuations. The control constraints are the pore pressure (lower constraint) and the fracture pressure (higher constraint). The margins towards the constraints should be high, since the pore- and fracture pressure are uncertain. In this paper, the control objective is to stay within ± 2 bar of the pressure set point.

3.3. First-order model approximation

The detailed model used for the simulations has been used in a nonlinear MPC in [9]. However, these detailed models need several parameters that are typically unknown in a real drilling operation.

A low-order model [21] fit for control purposes has been developed and used in control applications for managed pressure drilling [6,7], and observers for downhole pressure [22]. However, the model has currently only been used in laboratory facility setups. In addition, the model requires certain parameters such as the drill-string frictional pressure drop, and at the current drilling rigs this parameter cannot easily be measured. The model is a one-phase model and does not include a model for gas.

For an industrial application the search for a sufficiently accurate control design is needed. Even the simplest control design based on a first order model has not been evaluated sufficiently. Experiences from other industries using MPC approaches indicate that a first order model is often sufficient. In this paper, first-order step response models are therefore used in the controller design, similar to what is applied in [5].

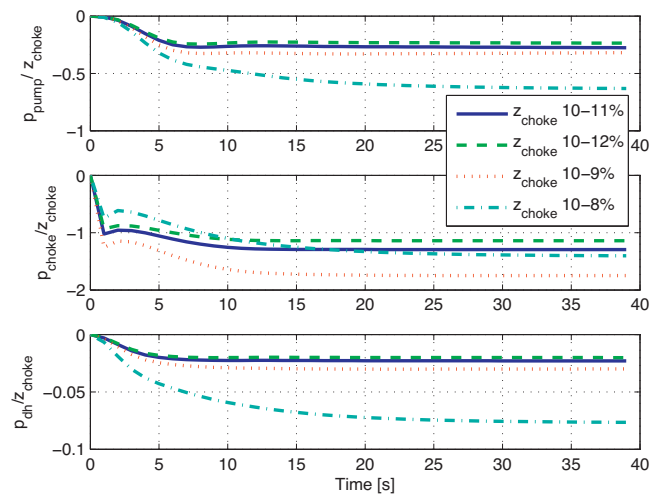


Fig. 3. Pressure step responses to various step changes in choke opening. Constant flow rate 1000 l/min.

The pressure step response when exposed to a step change in the choke opening or pump flow rate can be approximated by

$$p(s) = \frac{K_c}{\tau_c s + 1} z_c(s) + \frac{K_p}{\tau_p s + 1} q_p(s) \quad (1)$$

where p is the pressure, z_c is the choke opening, q_p is the pump rate, K_c and K_p are the process gains, and τ_c and τ_p are the time constants. The pressure p can be one of each pressure: choke pressure, pump pressure or downhole pressure. The process is non-linear, since the process gains K_c and K_p , and time constants τ_c and τ_p are dependent on the choke opening and flow rate. The parameters increase with increasing flow rate and decreasing choke opening.

The simulation model is here used as reality in order to define the parameters in the first order models. The experiments to define the parameters of the first order model can also be performed in a real drilling operation.

Fig. 3 shows the pump pressure, choke pressure and downhole pressure step responses to various step changes in the choke opening, when the pump rate is kept constant at 1000 l/min. Fig. 4 shows the pressure response with step changes in the pump flow rate. In these cases, the choke opening is constant at 10%. An example of first order model approximations to the pressure step responses after a step change in the choke opening is shown in

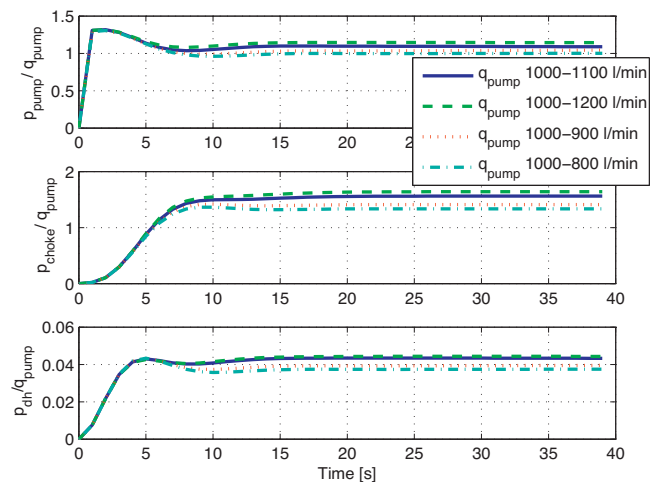


Fig. 4. Pressure step responses to various step changes in pump flow rate. Constant choke opening 10%.

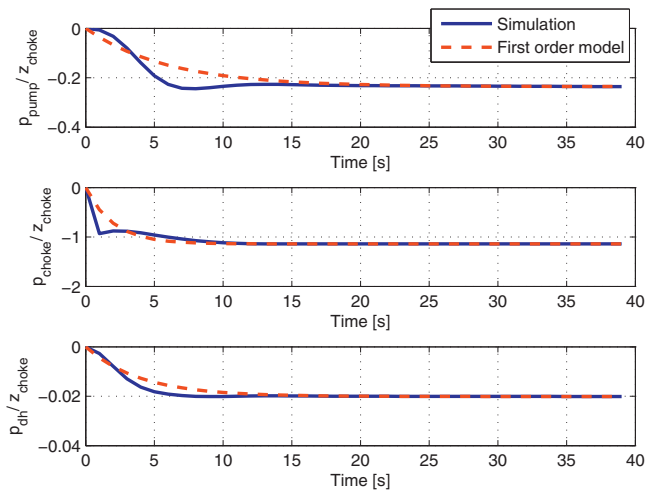


Fig. 5. Pressure step response to step change in choke opening from 10% to 12%. Blue solid line is from simulations, and red dashed line is a first order model approximation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

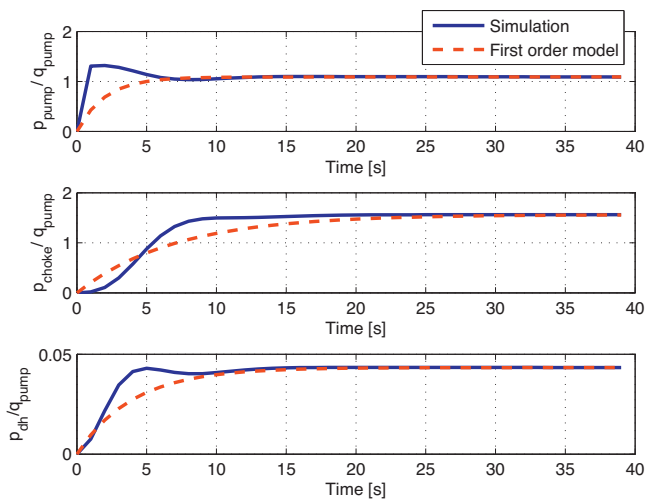


Fig. 6. Pressure step response to step change in flow rate from 1000 to 1100 l/min. Blue solid line is from simulations, and red dashed line is a first order model approximation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

Fig. 5. Fig. 6 shows first order model approximations to the pressure step responses after a step change in the pump flowrate.

4. Control schemes setup

Three different control methods have been evaluated for pressure control during a well control situation. The first is a standard PI control algorithm that controls the pressure by manipulating the choke opening. The second method is an IMC algorithm [23] controlling the pressure by manipulating the choke opening and the pump flowrate, using the first-order model of the well system from Eq. (1). The IMC structure controls pressure by adjusting the choke using a PI controller. Pump flow rate is controlled close to its setpoint using an integral-only controller. The entire control scheme is essentially a feedback/feedforward loop, using the choke as feedback manipulation and the pump flow rate as measured disturbance for feedforward corrections. The third control method is a linear MPC method [24]. This controller applies coordinated control of both the choke opening and the pump rate to control the pressure. The first-order model from Eq. (1) is used in the plant

Table 2
Controllers.

Controller type	Manipulated variables	Controlled output
PI	Choke opening	Choke pressure, pump pressure or downhole pressure
IMC	Choke opening and pump rate	Choke pressure, pump pressure or downhole pressure
MPC	Choke opening and pump rate	Choke pressure, pump pressure or downhole pressure

Table 3
Well control sequences.

Part of sequence	Controlled output without downhole measurement	Controlled output with downhole measurement
Shut in	Open loop	Open loop
Circulation start-up	Choke pressure	Downhole pressure
Circulation	Pump pressure	Downhole pressure

model. All three controllers have been developed for controlling choke pressure, pump pressure or downhole pressure. A list of the controllers and control variables is given in Table 2.

The three main parts of the well control sequence are well shut-in, pump start-up and circulation. In each of these parts the controlled output is different. In the well shut-in sequence the choke valve and main pump are operated using only open loop control. The main pump is ramped down at a pre-defined schedule and the valve is as a main rule closed when the pump is fully stopped. In the pump start-up sequence the controlled output is the choke pressure, and during the circulation sequence the controlled output is the pump pressure. For the IMC and the MPC controller the pump flowrate is a manipulated variable in addition to the choke valve opening. If the downhole pressure is available during the well control sequence then the controlled output is downhole pressure. An overview of the various sequences and the controlled output for each sequence is given in Table 3. Table 4 shows the form of the controllers, and the control values for each controller type and controlled output are given in Table 5. The tuning of the controllers is based on simulated response tests for each controller structure. A further discussion on the controller tuning is presented in Section 6.

5. Control schemes evaluation

The three controllers presented in the previous section have been evaluated for automatic pressure control during a well control procedure by running simulations using the detailed fluid model and the well test case described in Section 3. The lower pressure constraint is the pore pressure, and the upper constraint is fracture pressure at the casing shoe. However, the pore and fracture pressure in real operations are not known exactly, so the controller should always stay as close to the reference value as possible. The control objective is therefore to keep the pressure within ± 2 bar of the reference pressure during the procedure.

In Simulation 1, a kick incident occurs during a conventional drilling operation with no downhole pressure measurements available. The well is shut in and the kick is circulated out with automatic control of the choke and pump pressure. The simulation setup in Simulation 2 and Simulation 3 is identical to the first, except the pore pressure and permeability of the reservoir is increased respectively, in order to test the controller robustness. In Simulation 4, measurements of the downhole pressure are available, and the kick is circulated out while controlling the downhole pressure explicitly.

Table 4
Controller form.

Controller type	Controller form
PI	$C(s) = K_p \left(1 + \frac{1}{\tau_i s} \right)$
IMC	$C(s) = \begin{bmatrix} \frac{\tau_c}{K_c \lambda_1} \left(1 + \frac{1}{\tau_c s} \right) & -\frac{\tau_c}{K_c \lambda_2} \left(1 + \frac{1}{\tau_c s} \right) \frac{K_p}{\tau_p s + 1} \\ 0 & \frac{1}{\lambda_2 s} \end{bmatrix}$
MPC	$\min_{\Delta u(k), \dots, \Delta u(m-1+k)} \left\{ \sum_{i=1}^p \left[w(y(k+i) - y^{sp}) ^2 + r_1 \Delta u_1(k+i) ^2 + r_2 \Delta u_2(k+i) ^2 + v_1 (u_1(k+i) - u_{1target}(k+i)) ^2 + v_2 (u_2(k+i) - u_{2target}(k+i)) ^2 \right] \right\}$ s.t. $y_{min} \leq y(k+i) \leq y_{max}$

5.1. Simulation 1: well control using topside pressure measurements

This section presents results from simulations of a well control procedure where the downhole pressure measurements are not available, which is the case in most drilling operations. The procedure uses the choke and pump pressure measurements during circulation start-up and while circulating out the kick, as explained for the standard well control procedures in Section 2, only with automatic instead of manual control of the pressure.

Initially, a conventional drilling operation in the reservoir zone is going on. The pore pressure at the bottom of the hole is 262 bar. The blow-out preventer is open and the well control choke is closed. The pump is running at 2000 l/min, pumping drilling mud down the drill string, up the annulus and out through the open top of the annulus into atmospheric pressure. This pump rate gives a downhole pressure of 258 bar which is below the critical pore pressure at 262 bar, resulting in a situation where reservoir gas starts to flow into the bottom of the well. When the gas flows into the annulus, filling the fixed volume, the mud flow rate out of the annulus is higher than the mud flow rate pumped into the drillstring, and the pit tank volume will then start to increase. When the pit gain has increased above 2 m³, this is defined as a kick situation, and the well control procedure with automatic pressure control is initiated.

The choke pressure, pump pressure and downhole pressure during the whole simulation is shown in Fig. 7, and the time period for various sequences in the simulation are listed in Table 6.

Figs. 7–10 show the results from simulations where the PI, IMC, and MPC controllers are used to control the pressure, plotted in solid green, dashed red and dashdotted blue, respectively. The pore pressure is shown in dotted magenta. The PI controller uses the choke opening to control the choke pressure or pump pressure, and the IMC and MPC controllers use both the choke opening and pump rate to control the pressure during the well control procedure.

The pit gain reaches the set limit of 2 m³ after about 200 s. The well control procedure then starts by shutting in the well by

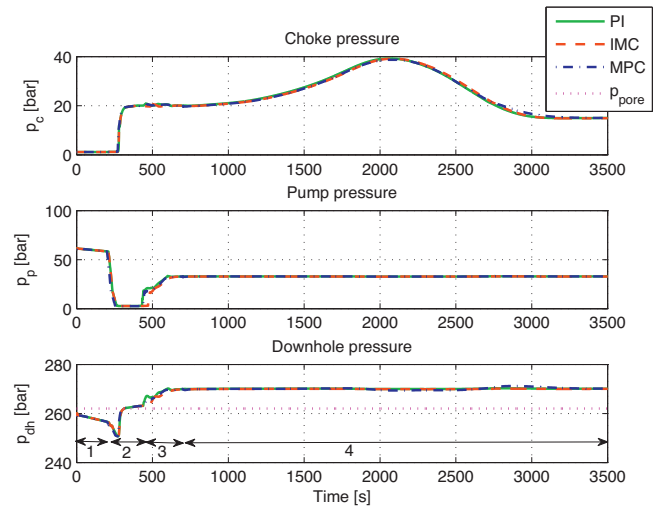


Fig. 7. Simulation 1: choke pressure, pump pressure, and downhole pressure and pore pressure while taking a kick (1), shutting in the well (2), starting the pump (3) and circulating out the gas kick (4).

closing the BOP, shutting down the pump and closing the well control choke.

After the well has been shut in, the pressure at bottom hole increases, as depicted in the upper plot in Fig. 8 from 270 to 300 s. This is because as long as the pressure at bottom is below the pore pressure, gas will continue to flow into the closed well, which causes the pressure increase. The gas influx gradually reduces as

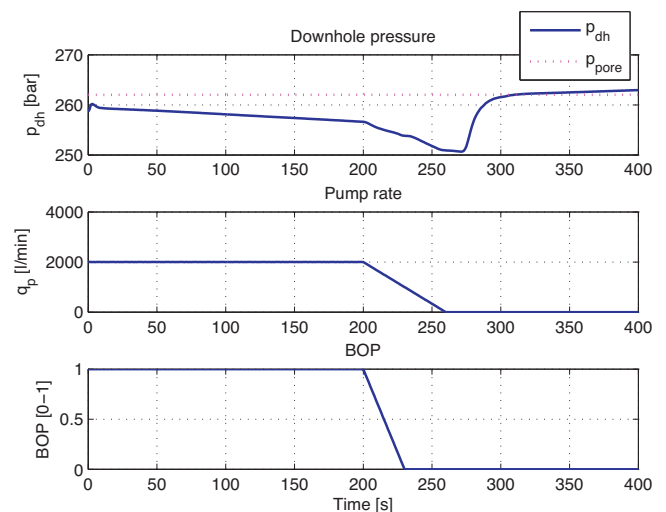


Fig. 8. Simulation 1: taking a kick and shutting in the well. Figures from top to bottom: downhole pressure (solid blue) and pore pressure (dotted magenta), pump flowrate, and BOP opening. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

Table 5
Controller parameters.

Controller type	Controlled output		
	Choke pressure	Pump pressure	Downhole pressure
PI	$K_p = -0.068$ $\tau_1 = 0.4$	$K_p = -2.3$ $\tau_1 = 10$	$K_p = -25$ $\tau_1 = 8$
IMC	$\lambda_1 = 0.86$ $\lambda_2 = 0.95$	$\lambda_1 = 0.83$ $\lambda_2 = 0.95$	$\lambda_1 = 0.7$ $\lambda_2 = 0.9$
MPC	$p = 100$ $m = 2$ $w = 0.85$ $r_1 = 0.1$ $r_2 = 0.1$ $v_1 = 0.0$ $v_2 = 0.001$	$p = 100$ $m = 2$ $w = 0.3$ $r_1 = 0.1$ $r_2 = 0.1$ $v_1 = 0.0$ $v_2 = 0.0015$	$p = 100$ $m = 2$ $w = 0.9$ $r_1 = 0.1$ $r_2 = 0.1$ $v_1 = 0.0$ $v_2 = 0.003$

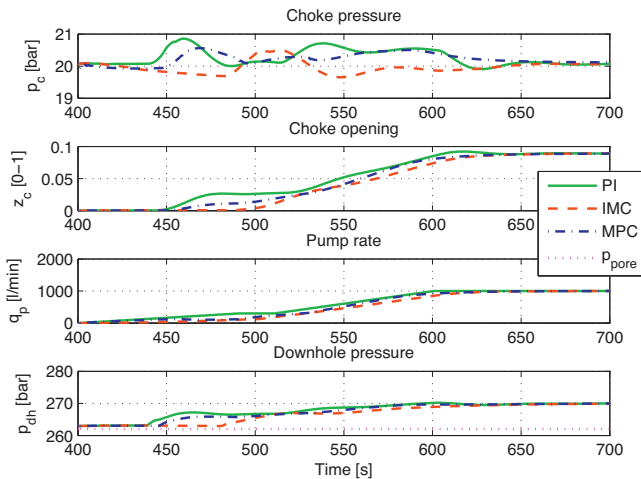


Fig. 9. Simulation 1: choke pressure, choke opening, pump rate, and downhole pressure and pore pressure while starting up the pump after shut-in, controlling the choke pressure.

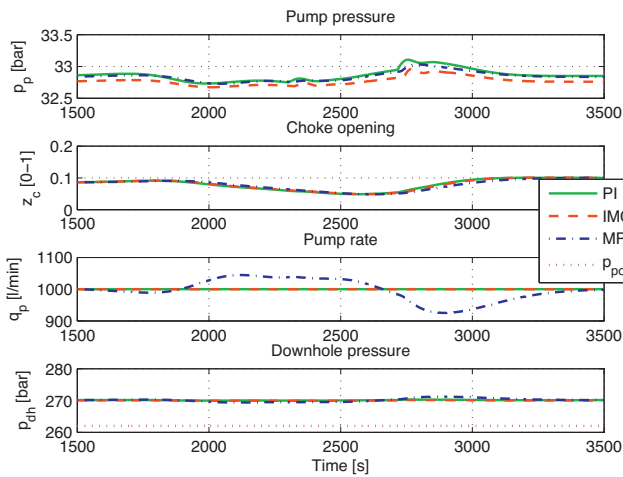


Fig. 10. Simulation 1: pump pressure, choke opening, pump rate, and downhole pressure and pore pressure while the gas influx is circulating out of the choke, controlling the pump pressure.

the downhole pressure increases towards the pore pressure. When the downhole pressure has increased above the pore pressure the pressure stabilizes, as can be seen in Fig. 8 after 300 s.

When the influx is stopped and the pressure has stabilized, the next step in the well control procedure is to start up the pump in order to circulate out the gas that has entered the annulus. According to the procedure, the choke pressure should be kept constant during pump start-up. In the simulations, the automatic choke pressure controller (PI, IMC or MPC) is activated during pump startup, at time 400 s. The PI controller manipulates only the choke opening in order to control the pressure. The pump is then ramped up from 0 to 1000 l/min using sequence logic. The IMC and MPC controllers manipulate both the pump and the choke opening to control the choke pressure, with the target pump rate set to 1000 l/min.

Table 6
Automatic well control simulation.

Sequence	Description	Time period (s)
1	Influx	0–200
2	Shut in well	200–400
3	Start circulation	400–700
4	Circulate influx	700–3500

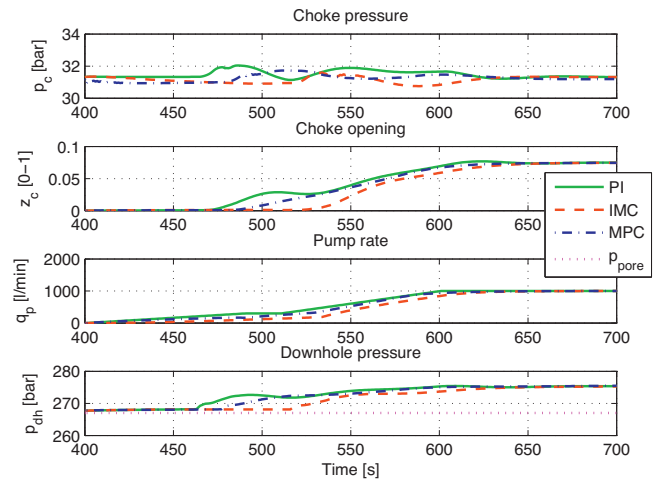


Fig. 11. Simulation 2: choke pressure, choke opening, pump rate, and downhole pressure and pore pressure while starting up the pump after shut-in controlling the choke pressure.

Fig. 9 shows the choke pressure, choke opening, pump rate, and downhole pressure and pore pressure during the pump startup sequence. The simulation using the PI controller is shown in solid green, the IMC in dashed red, and the MPC in dashdotted blue. All three controllers manage to control the choke pressure within ± 2 bar of the reference pressure. The reference pressure is the choke pressure at time 400 s, which is slightly different in the simulations of the different controllers.

When the pump is circulating at the target rate 1000 l/min after about 700 s, the procedure continues by circulating the kick out of the annulus. During this period, the pump pressure should be kept constant, so the controlled output is switched to pump pressure. Again, the PI controller manipulates only the choke opening, whereas the IMC and MPC use coordinated control of the choke opening and the pump rate. Fig. 10 shows the pump pressure, choke opening, pump rate, and downhole pressure during the time period when the gas exits the well control choke. All three controllers control the pump pressure within 0.5 bar of their reference pressure.

5.2. Simulation 2: increased reservoir pore pressure

To evaluate the robustness of the controllers, simulations have been run with a pore pressure of 267 bar instead of 262 bar in the reservoir section. The remaining simulation parameters are equivalent to Simulation 1. Results are shown in Figs. 11 and 12. The amount of gas entering the wellbore is doubled compared to when the pore pressure was 262 bar, when the time before the well is shut in remains the same.

Fig. 11 shows the choke pressure, choke opening, pump rate, and downhole pressure and pore pressure during the pump startup sequence after the well has been shut in. The controlled output is the choke pressure, and the controllers are trying to maintain a constant choke pressure. All three controllers have the same or lower pressure variations compared to the previous simulation where the pore pressure was lower, even though there is more gas causing pressure disturbances in the system. This can be explained by the fact that since more gas has entered the well, the choke pressure after shut-in is higher during this simulation. When the pressure is higher, changes in choke opening will have a higher effect on the pressure, so the control response is faster and more aggressive in this case. This may also indicate a nonlinearity in the valve characteristics, and improved controller tuning could be investigated in a future study.

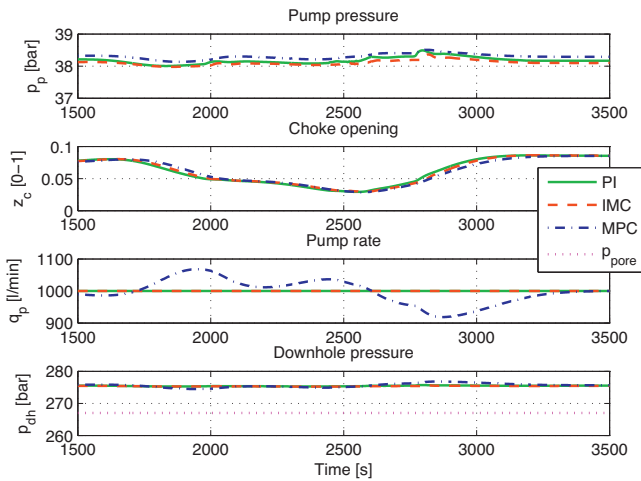


Fig. 12. Simulation 2: pump pressure, choke opening, pump rate, and downhole pressure and pore pressure while the gas influx is circulating out of the choke controlling the pump pressure.

Fig. 12 shows the pump pressure, choke opening, pump rate, and downhole pressure and pore pressure at the time when the gas exits the choke. The controlled output is the pump pressure, and the controllers are trying to maintain a constant pump pressure. The reference pump pressure is higher during this simulation, since the downhole pressure must remain above the pore pressure, and the pore pressure is higher in this simulation. The controllers are a bit more aggressive in this case, both due to the higher pressure and that the higher amount of gas is causing more disturbances. All controllers are within the 2 bar pressure window.

5.3. Simulation 3: increased reservoir permeability

To evaluate further the robustness of the controllers, simulations have been run where the permeability in the reservoir section is 400 mD instead of 100 mD, causing the gas to flow at a higher rate into the wellbore. The pore pressure is 262 bar, the same as in Simulation 1. The results are shown in Figs. 13 and 14.

During this simulation the amount of gas entering the well is almost four times as high compared to Simulation 1 where the permeability was lower. Fig. 13 shows the choke pressure, choke opening, pump rate, and downhole pressure and pore pressure

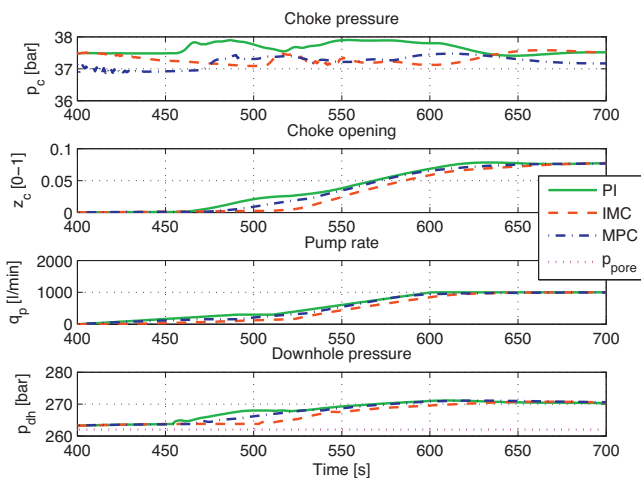


Fig. 13. Simulation 3: choke pressure, choke opening, pump rate, and downhole pressure and pore pressure while starting up the pump after shut-in controlling the choke pressure.

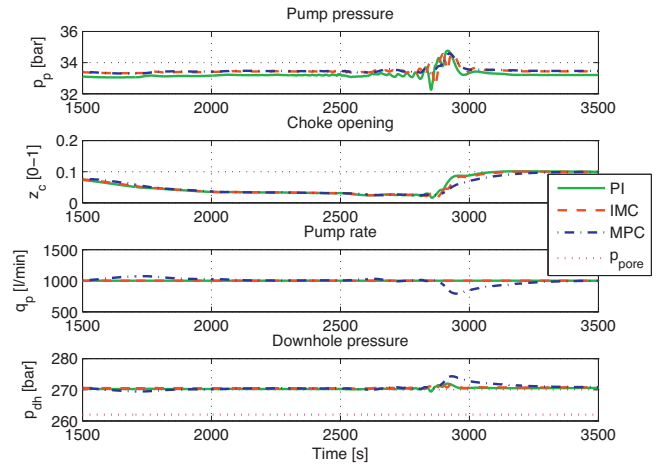


Fig. 14. Simulation 3: pump pressure, choke opening, pump rate, and downhole pressure and pore pressure while the gas influx is circulating out of the choke controlling the pump pressure.

during the pump startup sequence after the well has been shut in. The controlled output is the choke pressure, and the controllers are trying to maintain a constant choke pressure. Also during these simulations the pressure variations are slightly less, compared to the simulation where the permeability was lower, even though there is more gas causing pressure disturbances in the system. Again, this can be explained by the fact that since more gas has entered the well, the choke pressure after shut-in is higher during this simulation, resulting in faster control response. But in this case the model based controllers are more aggressive, as tuned, causing more oscillations in the pressure. A further tuning of the possible nonlinearity in the valve characteristics could be investigated in a future study.

Fig. 14 shows the pump pressure, choke opening, pump rate, and downhole pressure and pore pressure at the time when the gas exits the choke. The controlled output is the pump pressure, and the controllers are trying to maintain a constant pump pressure. In this simulation, the reference pump pressure is almost the same as for Simulation 1, since the pore pressure is the same. The pressure variations are higher during these simulations for all the controllers, since there is much more gas causing disturbances in the pressure. But all the controllers stay well within the control criteria of 2 bar pressure variations.

5.4. Simulation 4: well control using downhole pressure measurements

This section presents results from simulations where the downhole pressure measurements are available during the well control procedure. The pore pressure and permeability are the same as in Simulation 1. The simulation follows the same steps as Simulation 1, only the downhole pressure is now directly controlled and the PI, IMC and MPC controllers have the downhole pressure as controlled output during pump startup and while circulating out the kick.

Fig. 15 shows the downhole pressure and pore pressure, the choke opening, and the pump rate during the pump startup sequence. Fig. 16 shows the downhole pressure and pore pressure, the choke opening, and the pump rate while circulating out the kick. All the controllers manage to control the pressure within the 2 bar pressure window.

Fig. 17 shows a comparison of the simulations with the IMC controller with and without downhole pressure measurement, during pump startup and while circulating out the kick. The controller using topside measurements is shown in dashed red, and the controller using downhole measurements is shown in dashdotted blue.

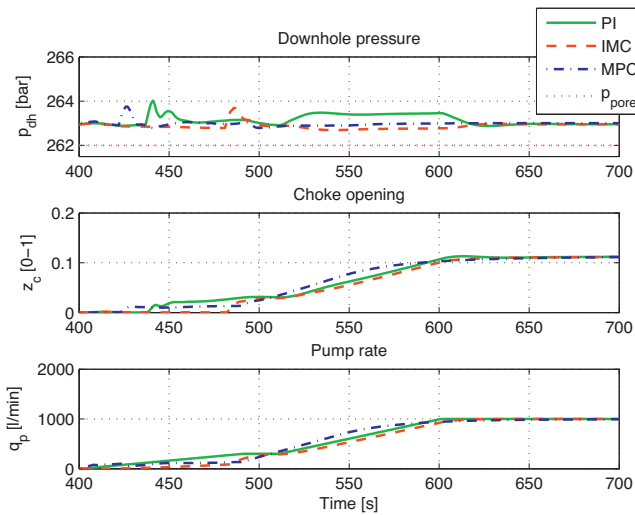


Fig. 15. Simulation 4: downhole pressure and pore pressure, choke opening, and pump rate while starting up the pump after shut-in controlling the downhole pressure.

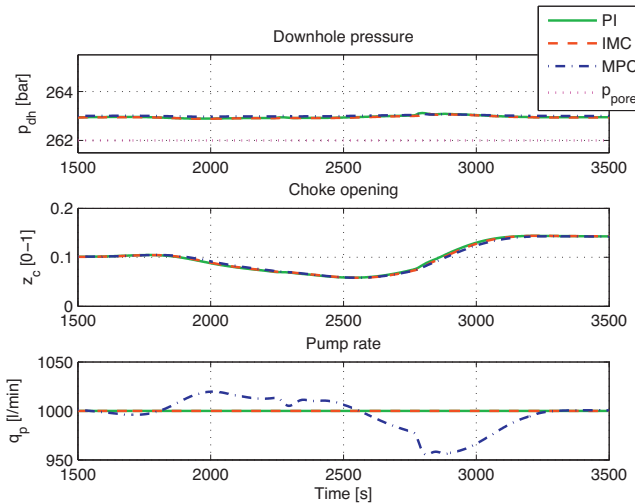


Fig. 16. Simulation 4: downhole pressure and pore pressure, choke opening, and pump rate while the gas influx is circulating out of the choke controlling the downhole pressure.

During pump startup (from 400 to 700 s), the IMC using topside measurements controls the choke pressure. As can be seen from the dashed red line in the upper plot in Fig. 17, the downhole pressure then increases from 263 to 270 bar. The pressure increase is due to increased frictional pressure loss in the annulus when starting the pump and increasing the flowrate. This pressure increase is considered as an extra safety margin towards the pore pressure. When controlling the downhole pressure directly the pressure remains constant during pump start-up, as can be seen from the dashdotted blue line in the upper plot in Fig. 17.

To examine what happens when the controllers are less aggressive, a simulation was run with a pore pressure at 259 bar and where less gas enters the wellbore before the well is shut in. The IMC using topside measurements and the IMC using the downhole pressure measurement were evaluated for this case, and the results are shown in Fig. 18 during pump startup and while circulating out the kick. Both controllers manage to control the pressure even though the controllers are less aggressive in this case. During pump startup, a slower control response will give a higher margin towards

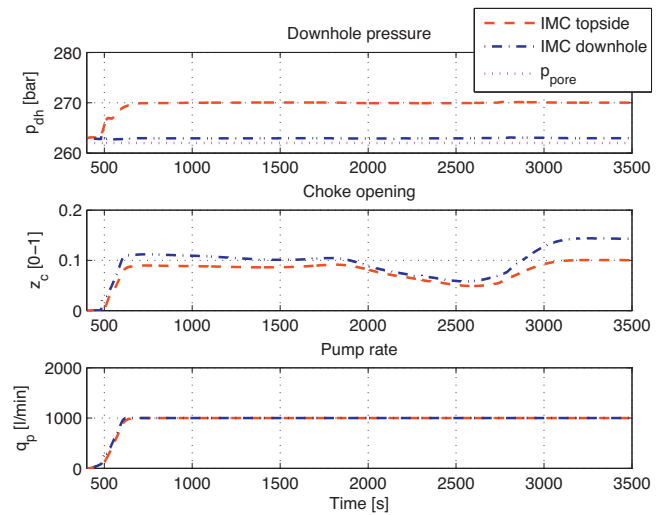


Fig. 17. Comparison of IMC controller with and without downhole pressure measurement. From top to bottom: downhole pressure and pore pressure, choke opening and pump flowrate. Red dashed lines: simulation using the IMC controller with only topside measurements available (choke pressure and pump pressure). Blue dashdotted lines: simulation using the IMC controller with downhole pressure measurements. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

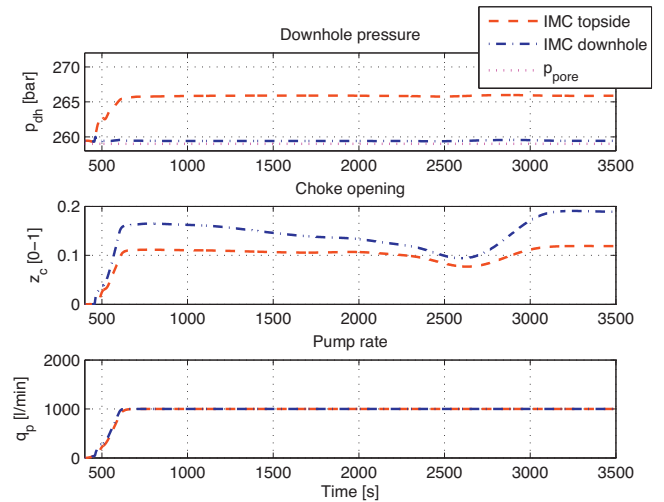


Fig. 18. IMC controller with and without downhole pressure measurement when the pore pressure is 259 bar. From top to bottom: downhole pressure and pore pressure, choke opening and pump flowrate. Red dashed lines: simulation using the IMC controller with only topside measurements available (choke pressure and pump pressure). Blue dashdotted lines: simulation using the IMC controller with downhole pressure measurements. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

the pore pressure since the choke will open more slowly when the pump is ramped up.

6. Discussion on controller tuning

The various controllers used in the simulations have been tuned by looking at the dynamic response of the wellbore. The PI and the IMC controllers have few parameters and the selection of parameters has been relatively straight forward. Regarding the MPC controller, a larger set of parameters has been examined. A special focus has been on defining the best control horizon. Control performance improves by increasing the control horizon, but robustness is reduced as m becomes larger.

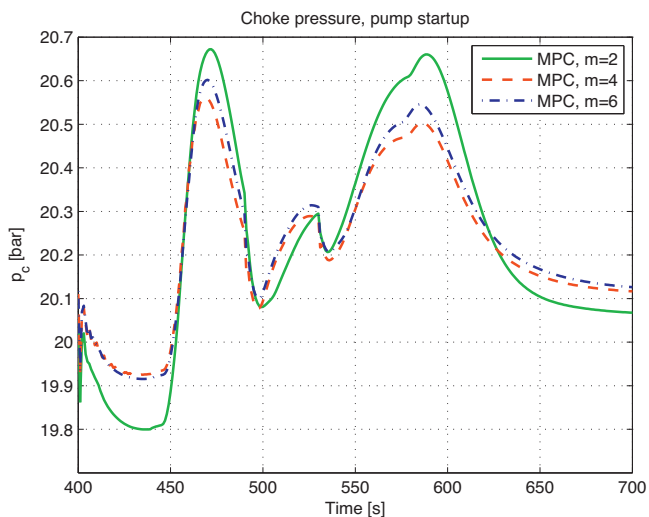


Fig. 19. MPC tuning, control horizon: choke pressure while starting up the pump after shut-in controlling the choke pressure using MPC control with various control horizon values m . Pore pressure at 262 bar.

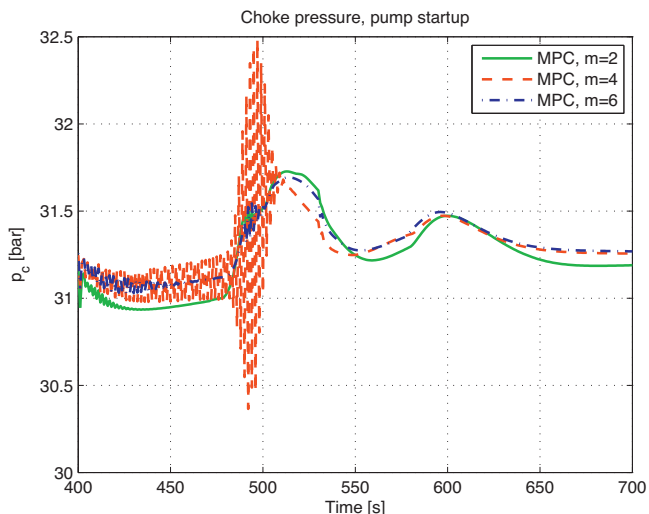


Fig. 20. MPC tuning, control horizon: choke pressure while starting up the pump after shut-in controlling the choke pressure using MPC control with various control horizon values m . Pore pressure at 267 bar.

Fig. 19 shows the choke pressure during pump startup when using the MPC controller to control the pressure with various values for the control horizon m ; $m = 2$, $m = 4$ and $m = 6$. The pore pressure is 262 bar, as in Simulation 1. Fig. 20 shows the results when running the simulation with a pore pressure at 267 bar, as in Simulation 2. In this case, there are more pressure oscillations with larger values for the control horizon m . With larger control horizon, the MPC controller tries to be as close as possible to the inverse of the built-in linear model. When the pore pressure is higher, more gas has entered the wellbore before shut-in. The linear model in the MPC controller is not accurate with the large amount of gas, and the resulting control action is too aggressive, thus creating the observed oscillations. We have found the optimal control horizon in this case to be $m = 2$.

7. Conclusions and further work

The simulations show that it is feasible to control the pressure using automatic control of the choke valve and pump during an

unexpected gas influx by use of the presented control methods. The control methods are robust against changes in process conditions and disturbances, as they are able to handle several pressure levels and gas volumes without requiring re-tuning. However, since the pressure dynamics in the well is influenced when gas is entering the well, the model based controllers could probably be improved if the models were updated after the gas influx occurred.

The results indicate that adaption of the automatic sequence to the current manual procedure is applicable. However, to avoid a reduction in downhole pressure when stopping the pump and shutting in the well, the automatic sequence may be further improved beyond what is feasible with manual operation.

The current methods show that automatic pressure control while gas is present in the well is possible to achieve. Underbalanced drilling (UBD), where the well pressure is below the pore pressure during the whole operation, is closely related to the well control operation. The pressure control during UBD operations is currently performed manually, but this operation could be automated. The path towards automation of the UBD operation and the well control operation should be followed in close relation.

Two different paths should be followed regarding the future work on the topic on automatic well control for drilling operations. Firstly, there should be an investigation how the simplified models may be updated after the gas influx occurs. This could lead to a further improved performance of the model based controllers. The pore pressure and permeability range within which the controllers are robust could also be examined more thoroughly. In the simulations it was only examined that the downhole pressure did not fall below the pore pressure while circulating the kick. The upper pressure constraint is the fracture pressure of the formation at the casing shoe, which could be investigated further in future work.

The second path that should be followed regarding future work is the shut-in sequence. The current shut-in sequence is only using sequence logic and gives reduced downhole pressure. From the simulations we see that this leads to increased influx. Therefore, an improved shut-in control sequence utilizing both pump flow rate control and choke valve pressure control should be investigated.

Acknowledgements

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References

- [1] R.D. Grace, *Blowout and Well Control Handbook*, Elsevier Science, Burlington, MA, 2003.
- [2] K. Malloy, C. Stone, J.G.H. Medley, D. Hannegan, O. Coker, D. Reitsma, H. Santos, J. Kinder, J. Eck-Olsen, J. McCaskill, J. May, K. Smith, P. Sonnemman, Managed-pressure drilling: what it is and what it is not, in: IADC/SPE Managed Pressure Drilling and Underbalanced Operations Conference and Exhibition, San Antonio, TX, 2009.
- [3] D. Reitsma, Development and application of combining a real-time hydraulics model and automated choke to maintain a relatively constant bottomhole pressure while drilling, in: International Petroleum Technology Conference, Doha, Qatar, 2005.
- [4] D. Reitsma, T. Runggai, N. Hudson, R. Zaeper, O. Backhaus, M. Hernandez, Successful implementation of first closed loop, multiservice control system for automated pressure management in a shallow gas well offshore Myanmar, in: SPE/IADC Drilling Conference, Amsterdam, The Netherlands, 2008.
- [5] J. Godhavn, Control requirements for high-end automatic mpd operations, in: SPE/IADC Drilling Conference and Exhibition, Amsterdam, The Netherlands, 2009.
- [6] J. Zhou, G. Nygaard, Control and estimation of downhole pressure in managed pressure drilling operations, in: Proceedings of the 4th International Symposium on Communications, Control and Signal Processing, Limassal, Cyprus, 2010.

- [7] J.-M. Godhavn, A. Pavlov, G.-O. Kaasa, N.L. Rolland, Drilling seeking automatic control solutions, in: Proceedings of the 18th IFAC World Congress, Milano, Italy, 2011.
- [8] Ø. Breyholtz, G. Nygaard, M. Nikolaou, Advanced pressure control for dual-gradient drilling, in: SPE Annual Technical Conference and Exhibition, New Orleans, LA, 2009.
- [9] G. Nygaard, G. Nævdal, Non-linear model predictive control scheme for stabilizing annulus pressure during oil well drilling, *Journal of Process Control* 16 (7) (2006) 719–732.
- [10] Ø. Breyholtz, G. Nygaard, M. Nikolaou, Managed pressure drilling: a multi-level control approach, in: SPE Intelligent Energy Conference and Exhibition, Utrecht, The Netherlands, 2010.
- [11] Ø. Breyholtz, G. Nygaard, M. Nikolaou, Automatic control of managed pressure drilling, in: Proceedings of American Control Conference, Baltimore, USA, 2010.
- [12] M. Davoudi, J. Smith, B. Patel, J. Chirinos, Evaluation of alternative initial responses to kicks taken during managed pressure drilling, in: IADC/SPE Drilling Conference and Exhibition, New Orleans, LA, USA, 2010.
- [13] H. Santos, P. Reid, J. Jonesand, J. McCaskill, Developing the micro-flux control method-part 1: system development field test preparation and results, in: SPE/IADC Middle East Drilling Technology Conference and Exhibition, Dubai, United Arab Emirates, 2005.
- [14] M. Stanislawek, J.R. Smith, Analysis of alternative well-control methods for dual-density deepwater, in: IADC/SPE Drilling Conference, Miami, FL, USA, 2006.
- [15] J. Zhou, G. Nygaard, Automatic model-based control scheme for stabilizing pressure during dual-gradient drilling, *Journal of Process Control* 21 (8) (2011) 1138–1147.
- [16] J. Zhou, G. Nygaard, J. Godhavn, Ø. Breyholtz, E. Vefring, Adaptive observer for kick detection and switched control for bottomhole pressure regulation and kick attenuation during managed pressure drilling, in: Proceedings of American Control Conference, Baltimore, USA, 2010.
- [17] J. Gravdal, M. Nikolaou, Ø. Breyholtz, L. Carlsen, Improved kick management during mpd by real-time pore-pressure estimation, in: SPE Annual Technical Conference and Exhibition, New Orleans, LA, 2009.
- [18] R.J. Lorentzen, G. Nævdal, A. Lage, Tuning of parameters in a two-phase flow model using an ensemble kalman filter, *International Journal of Multiphase Flow* 29 (2003) 1284–1309.
- [19] E. Cayeux, B. Daireaux, Early detection of drilling conditions deterioration using real-time calibration of computer models: field example from north sea drilling operations, in: Proceedings of the SPE/IADC Drilling Conference and Exhibition, Amsterdam, The Netherlands, 2009.
- [20] E. Cayeux, B. Daireaux, E. Dvergsnes, G. Sælevik, M. Zidan, An early warning system for identifying drilling problems: an example from a problematic drill-out cement operation in the north-sea, in: Proceedings of the SPE/IADC Drilling Conference and Exhibition, San Diego, CA, 2012.
- [21] G.-O. Kaasa, A Simple Dynamic Model of Drilling for Control, Tech. rep., Statoil-Hydro, Porsgrunn, Norway, 2007.
- [22] Ø.N. Stamnes, J. Zhou, G. Kaasa, O. Aamo, Adaptive observer design for the bottomhole pressure of a managed pressure drilling system, in: Proceedings of the 47th IEEE Conference on Decision and Control, Cancun, Mexico, 2008.
- [23] M. Morari, E. Zafriou, Robust Process Control, Prentice-Hall, USA, 1989.
- [24] J. Maciejowski, Predictive Control with Constraints, Pearson Education, Essex, England, 2002.